3. Prove that the product of two consecutive positive integers is divisible by 2.
4. When the positive integers a, b and c are divided by 13, the respective remainders are 9,7 and 10. Show that $a+b+c$ is divisible by 13.
5. Prove that square of any integer leaves the remainder either 0 or 1 when divided by 4.
6. Use Euclid's Division Algorithm to find the Highest Common Factor (HCF) of
   (i) 340 and 412 (ii) 867 and 255
   (iii) 10224 and 9648 (iv) 84, 90 and 120
7. Find the largest number which divides 1230 and 1926 leaving remainder 12 in each case.
8. If d is the Highest Common Factor of 32 and 60, find x and y satisfying $d = 32x + 60y$.
9. A positive integer when divided by 88 gives the remainder 61. What will be the remainder when the same number is divided by 11?
10. Prove that two consecutive positive integers are always coprime.

2.4 Fundamental Theorem of Arithmetic

Let us consider the following conversation between a Teacher and students.

Teacher : Factorise the number 240.
Malar : 24 ×10
Raghu : 8×30
Iniya : 12×20
Kumar : 15×16
Malar : Whose answer is correct Sir?
Teacher : All the answers are correct.
Raghu : How sir?
Teacher : Split each of the factors into product of prime numbers.
Malar : 2×2×2×3×2×5
Raghu : 2×2×2×2×3×5
Iniya : 2×2×3×2×2×5
Kumar : 3×5×2×2×2×2
Teacher : Good! Now, count the number of 2's, 3's and 5's.
Malar : I got four 2's, one 3 and one 5.
Raghu : I got four 2's, one 3 and one 5.
Iniya : I also got the same numbers too.
Kumar : Me too sir.
Malar : All of us got four 2's, one 3 and one 5. This is very surprising to us.
Teacher : Yes, It should be. Once any number is factorized up to a product of prime numbers, everyone should get the same collection of prime numbers.

This concept leads us to the following important theorem.

2 Numbers and Seq.indd 43 15-12-2020 02:52:40 PM